Question: You drive 200 miles in 3 hours before stopping for 30 minutes for lunch and gas. After lunch you travel 150 miles in an hour and a half. What was your average speed for the trip?
Answer: To find the average speed for the entire trip, you need to calculate the total distance traveled and divide it by the total time taken. Total distance = 200 miles + 150 miles = 350 miles. Total time = 3 hours + 0.5 hours + 1.5 hours = 4 hours. Average speed = Total distance / Total time = 350 miles / 4 hours = 70 mph.

User: You are a hallucination detection agent. Given a question and its corresponding response, please determine whether there is any incorrect or unsatisfactory content within the response. If there is, output "Yes"; if there is not, output "No".
Assistant: No.

simple prompt

Assistant: I will use the search engine tool to get relevant information.
web_search("To find the average speed for the entire trip, you need to calculate ..."

Retrieved Eesult: The average speed formula is: AverageSpeed(s) = total elapsed time / total distance.

There are no contradictions between the response and the retrieved fact.

No.

single tool

THOUGHT: I need to split the provided text into individual sentences to begin the hallucination detection process.
ACTION: split_text(text="To find the average speed for the entire trip, you need to ...").
OBSERVATION: ["To find the ...", "Total distance = ...", "Total time = ...", "Average speed = Total distance / Total time ..."]

THOUGHT: Fact-check the first sentence using the search tool to find relevant factual information.
ACTION: web_search(sentence="To find the average speed for the entire trip, you need to calculate the total distance traveled and divide it by the total time taken.")
OBSERVATION: "The average speed formula is: AverageSpeed(s) = total elapsed time / total distance."

THOUGHT: There are no contradictions between the first sentence and the retrieved fact, label = 0. Fact-check the second sentence using the calculator tool to get the answer to the formula.
ACTION: calculator(formula="200+150")
OBSERVATION: 200+150 = 350

THOUGHT: The result of the formula in the second sentence is right, label = 0. Fact-check the third sentence using the calculator tool to get the answer to the formula.
ACTION: calculator(formula="3+0.5+1.5")
OBSERVATION: 3+0.5+1.5 = 5

THOUGHT: The result of the formula in the third sentence is wrong, label = 1.
.....

THOUGHT: The result of the formula in the fourth sentence is right, label = 0. All test results are correct and do not require correction. Detection ended.
ACTION: get_answer()
OBSERVATION: Yes. { "sentence": "Total time = 3 hours + 0.5 hours + 1.5 hours = 4 hours.", "fact": "3+0.5+1.5=5"}

HaluAgent